ALGEBRA 2 · UNIT 6: EXPONENTIAL & LOGARITHMIC FUNCTIONS

Earthquake & Sound Lab

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students use logarithmic scales to compare earthquake amplitudes and sound intensities, model radioactive decay and cooling with exponential functions, solve exponential equations with logarithms, and design a public safety poster.

Log scales hide enormous differences: a magnitude 7 quake has 100 times the amplitude of a magnitude 5. Students also solve real exponential equations using logs.

Standards Addressed

- F-LE.4 Solve exponential equations using logarithms
- F-BF.5 Inverse relationship between exponents and logarithms
- F-IF.8b Interpret exponential growth and decay
- A-SSE.3c Transform exponential expressions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with earthquake news. Part 1 log scales.
Day 2	Part 2 decay and cooling models.
Day 3	Part 3 poster.

Materials

- Student handout
- Scientific calculators
- Optional: a free decibel-meter app

How the Three Versions Differ

Level	What changes
CP2	Evaluate logs of powers of 10, compare scales, and count half-lives.
CP1	Solve exponential equations with logs, log properties, and Newton's law of cooling.
Honors	Combine sound levels, build models from two points, and solve equations with different bases.

Watch For (Common Misconceptions)

- Thinking magnitude 7 is 2 times stronger than magnitude 5.
- log(a + b) = log a + log b.
- Using the half-life as a rate.

Extension Ideas

- Measure classroom sound levels.
- Research the largest earthquakes.
- Model a medicine's half-life.

Answer Key – CP2

Part 1: Log Scales

#	Task	Answer
1	A quake's amplitude is 100,000 times A_0. Find its magnitude.	log(100,000) = 5
2	How many times larger is the amplitude of a magnitude 7 quake than a magnitude 5 quake?	$10^7 \div 10^5 = 100$ times
3	A whisper is 30 dB and a concert is 110 dB. How many times more intense is the concert?	Difference 80 dB = 8 tens of dB: $10^8 = 100{,}000{,}000$ times
4	Evaluate log 1000, log 0.01, and log 1.	3, −2, 0

Part 2: Decay & Growth

#	Task	Answer
5	A bone has 25% of its original carbon-14. How many half-lives have passed? How old is it?	2 half-lives: 11,460 years
6	A bone has 12.5% left. How old is it?	3 half-lives: 17,190 years
7	\$1,000 grows continuously at 5% for 10 years: $A = 1000e^{0.5}$. Find A. Compare with monthly compounding: $1000(1 + 0.05 \div 12)^{120}$.	$1000e^{0.5} \approx$ \$1,648.72; monthly ≈ \$1,647.01 (continuous earns about \$1.71 more)

Part 3: Safety Poster

#	Task	Answer
8	Choose three sounds or quakes and compare them with log scales.	Answers vary.
9	Explain why each step of 10 dB means 10 times the intensity.	L increases by 10 when $I \div I_0$ is multiplied by 10, because 10 log(10x) = 10 + 10 log x.
10	Add one safety tip backed by your math.	Answers vary.

Answer Key – CP1

Part 1: Log Scales

#	Task	Answer
1	Compare the amplitudes of magnitude 6.8 and 5.3 quakes.	$10^{6.8-5.3} = 10^{1.5} \approx 31.6$ times
2	Each increase of 1 in magnitude multiplies the energy by about $10^{1.5} \approx 31.6$. About how many times more energy does a magnitude 7 release than a…	$31.6^2 \approx 1{,}000$ times
3	Condense log 2 + log 50 and expand $\log(100x^3)$.	log 100 = 2; 2 + 3 log x

Part 2: Solve with Logs

#	Task	Answer
4	A fossil has 30% of its carbon-14 left. Solve for its age with logarithms.	$0.3 = 0.5^n \rightarrow n = \log 0.3 \div \log 0.5 \approx 1.737$; $t = 5730n \approx 9{,}953$ years
5	Coffee cools by $T = 70 + 130e^{-0.08t}$ (°F, minutes). When will it reach 100°F?	$130e^{-0.08t} = 30 \rightarrow t = \ln(30 \div 130) \div (-0.08) \approx 18.33$ min
6	Solve $2^{x+3} = 50$.	$x + 3 = \log 50 \div \log 2 \approx 5.644 \rightarrow x \approx 2.644$

Part 3: Safety Poster

#	Task	Answer
7	Compare three events with log scales.	Answers vary.
8	Include one exponential model solved with logs.	Answers vary.
9	Write a safety message for the public.	Answers vary.

Answer Key – Honors

Part 1: Scales & Sums

#	Task	Answer
1	Two speakers each produce 80 dB. Find the combined level. Explain why it is not 160 dB.	$10 \log(2 \times 10^8) = 80 + 10 \log 2 \approx 83.0$ dB. Intensities add, not decibels.
2	How many 80-dB speakers would it take to reach 100 dB?	Need $10^2 = 100$ speakers.
3	Prove $\log(a \div b) = \log a - \log b$ from the exponent rules.	Let $a = 10^m$ and $b = 10^n$; $a \div b = 10^{m-n}$, so $\log(a \div b) = m - n = \log a - \log b$.

Part 2: Models

#	Task	Answer
4	A bacteria culture has 500 cells at t = 2 h and 4,500 at t = 6 h. Find a and b in $N = a \cdot b^t$, then write it as $N = ae^{kt}$.	$b^4 = 9 \rightarrow b = \sqrt{3} \approx 1.732$; $a = 500 \div 3 \approx 166.7$. $k = \ln \sqrt{3} \approx 0.549$: $N \approx 166.7e^{0.549t}$
5	Solve $3^{x+1} = 5^x$.	$(x + 1) \ln 3 = x \ln 5 \rightarrow x = \ln 3 \div (\ln 5 - \ln 3) \approx 2.151$
6	A fossil has 30% of its carbon-14. Find its age, then find the age if the measurement could be off by ±2 percentage points.	≈ 9,953 years; 28%: ≈ 10,523 years; 32%: ≈ 9,419 years

Part 3: Poster

#	Task	Answer
7	Include a model built from two data points.	Answers vary.
8	Include an error range for one calculation.	Answers vary.
9	Write a public message.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Comparisons and models are correct and explained in plain language for the public. Goes beyond the minimum with a creative, realistic choice.	Comparisons and models are correct and explained in plain language for the public.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Poster shows the log scales, at least three comparisons with work, and an exponential model solved with logarithms. Polished, organized, and visually engaging.	Poster shows the log scales, at least three comparisons with work, and an exponential model solved with logarithms. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.